



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MAT2060
Course title (English)	Honours Mathematical Analysis
Course title (Chinese)	数学分析荣誉课程
Units	3
Language of Instruction	English
Description (English)	This course is for students who have taken the courses Calculus Extended I and II (MAT1011 and MAT1012). This course is designed for outstanding students who are interested in mathematics or applied mathematics, and it aims to build a solid foundation of analysis for students. This course is a high-level course with high degree of difficulty, and it intends to cultivate innovative mathematics talents. The contents and teaching methods of this course emphasize comprehensive understanding of the theory and methods of mathematical analysis, and cultivate students' ability of in-depth analysis, rigorous reasoning, and courage to innovate. Topics include the systems of real and complex numbers, point set topology in Euclidean spaces, metric spaces, Baire category theorem, differentiation and integration, implicit function theorem and its application, sequences and series of functions, space of continuous functions, Fourier series, Fourier transform and Laplace transform, vector field theory and applications, Stokes' and Divergence Theorems.
Description (Chinese)	选修本门课程的学生须已修习了微积分精讲（一）与微积分精讲（二）（MAT1011和MAT1012）。本课程面向对数学或应用数学感兴趣的优秀学生，是一门高水平、高难度的课程，旨在为学生打下扎实的分析知识基础、培养数学创新人才。因此，本课程的教学内容和教学方法更强调对数学分析理论和方法的全面解析，培养学生深入分析、严谨推理、勇于创新的能力。其内容包括实数和复数、欧几里得空间的点集拓扑、度量空间、拜耳范畴定理、微分与积分、隐函数定理及其应用、函数序列与级数、连续函数空间、傅立叶级数、傅立叶变换与拉普拉斯变换、向量场理论与应用、Stokes定理与散度定理。

2. Prerequisites / Co-requisites



A. Prerequisites

MAT 1011 (Calculus I)

MAT 1012 (Calculus II)

B. Co-requisites

3. Learning Outcomes

Upon successful completion of the course, the students will have acquired the ability to (Knowledge, Skill, Value):

Knowledge:

- a) Students will understand real number system and topology in Euclidean spaces.
- b) Students will understand the theory of the space of continuous functions, the implicit function theorem and applications, and Riemann integrability.
- c) Students will understand Fourier series, Fourier transform and applications.
- d) Students will understand the vector field theory.

Skills:

- e) Apply the methods and techniques of this course to solving mathematical problems in analysis and to prove mathematical results in a logical and coherent fashion.
- f) Develop abstract concepts in advanced mathematics, and develop thought processes for the precision required in advanced mathematics.
- g) Develop ability of analyzing, computing and proving on a rigorous basis.

Value/Attitude:

- h) Students will understand the necessity of logical presentation and rigorous verification of mathematical results.
- i) Students will appreciate the scientific beauty of mathematical analysis and understand power of mathematical theory in solving problems.

4. Course syllabus

Topics to be covered:



- ☐ Real numbers, cardinality, Schroeder-Bernstein Theorem
- ☐ point set topology in Euclidean spaces
- ☐ Metric spaces, Baire category theorem
- ☐ Differentiation and integration
- ☐ Implicit function theorem and its application
- ☐ Sequences and series of functions, space of continuous functions
- ☐ Fourier series, Fourier transform and Laplace transform
- ☐ Vector field theory and applications, Stokes' and Divergence Theorems

5. Assessment Scheme

Component/ method	% weight
Homework	25
Midterm Exam	25
Final Exam	50

6. Grade descriptor

General

Grade	Description
A	Outstanding performance on all learning outcomes.
A-	Outstanding performance on all learning outcomes.
B	Substantial performance on all learning outcomes, or high performance on some learning outcomes which compensates for less satisfactory performance on others, resulting in overall substantial performance.
C	Satisfactory performance on the majority of learning outcomes, possibly with a few weaknesses.
D	Barely satisfactory performance on a number of learning outcomes
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

Collect student feedback from CTE by the end of the course

Collect opinions from peers by internal and external course reviews



8. Reading

Required Reading

1.A First Course in Real Analysis, 2nd edition, 2, United States

9. Course components

Activity	Hours/week
Lecture	3
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
00000 00001	Real numbers, cardinality, Schroeder-Bernstein Theorem
00000 00002	Point set topology in Euclidean spaces
00000 00003	Limits and continuity
00000 00004	Metric spaces
00000 00005	Baire category theorem and applications
00000 00006	Differentiation
00000 00007	Integration
00000 00008	Implicit function theorem and its application
00000 00009	Sequences and series of functions
00000 00010	Space of continuous functions, Ascoli-Arzelà theorem
00000 00011	Fourier series, Fourier transform and Laplace transform
00000 00012	Tietze extension theorem, Stone-Weierstrass theorem
00000 00013	Vector field theory and applications



00000	Stokes' and Divergence Theorems
00014	

11. Any other information

Prohibit the use of AI tools: AI tools (particularly GPT and generative AI tools) are not allowed in submitted reports. Any student who violates this rule will be considered to be engaging in academic dishonesty. The instructor may specify which AI tools are prohibited in this course.

12. Version date

Version number	3
As of (date)	2024/08/12